THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 10: Vector Algebra (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	The unit vector perpendicular to the vectors $\hat{\imath}-\hat{\jmath}$ and $\hat{\imath}+\hat{\jmath}$ is: (A) $\hat{k}$ (B) $-\hat{k}+\hat{\jmath}$ (C) $\frac{\hat{\imath}-\hat{\jmath}}{\sqrt{2}}$ (D) $\frac{\hat{\imath}+\hat{\jmath}}{\sqrt{2}}$	1
2.	If $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, $\lvert\vec{a}\rvert=\sqrt{37}$, $\lvert\vec{b}\rvert=3$ and $\lvert\vec{c}\rvert=4$, then the angle between $\vec{b}$ and $\vec{c}$ is: (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$	1
3.	The value of λ for which $2\hat{\imath}+\lambda\hat{\jmath}+\hat{k}$ and $\hat{\imath}-2\hat{\jmath}+3\hat{k}$ are perpendicular is: (A) $\frac{5}{2}$ (B) $-\frac{5}{2}$ (C) 2 (D) $\frac{3}{2}$	1
4.	If $\lvert\vec{a}\rvert=2$, $\lvert\vec{b}\rvert=5$ and $\vec{a}\cdot\vec{b}=8$, then $\lvert\vec{a}\times\vec{b}\rvert$ is: (A) 2 (B) 4 (C) 6 (D) 8	1
5.	The projection of $\vec{a}=2\hat{\imath}+3\hat{\jmath}+2\hat{k}$ on $\vec{b}=\hat{\imath}+2\hat{\jmath}+\hat{k}$ is: (A) $\frac{10}{\sqrt{17}}$ (B) $\frac{5\sqrt{6}}{3}$ (C) $\frac{10}{6}$ (D) $\sqrt{6}$	1

Q No.	Question	Marks
6.	**Assertion (A):** The vectors $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ and $\vec{b} = 2\hat{\imath} + 4\hat{\jmath} + 6\hat{k}$ are collinear. **Reason (R):** Two vectors $\vec{a}$ and $\vec{b}$ are collinear if $\vec{b} = \lambda\vec{a}$ for some scalar λ. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find the area of the parallelogram whose adjacent sides are $\vec{a} = \hat{\imath} + \hat{\jmath} - \hat{k}$ and $\vec{b} = \hat{\imath} - \hat{\jmath} + \hat{k}$.	2
	LEVEL 2 **Q 8 – Q 14**	**16**
8.	Find a vector of magnitude 21 units in the direction opposite to that of $\overrightarrow{AB}$, where A and B are the points $A(2,1,3)$ and $B(8,-1,0)$ respectively.	2
9.	Find a unit vector perpendicular to both $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$ and $\vec{b} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$.	2
10.	Two friends flying kites from different locations find the strings of their kites crossing each other. The strings can be represented by the vectors $\vec{a} = 3\hat{\imath} + \hat{\jmath} + 2\hat{k}$ and $\vec{b} = 2\hat{\imath} - 2\hat{\jmath} + 4\hat{k}$. Determine the angle formed between the kite strings. (Assume there is no slack in the strings.)	2
11.	Find the area of the triangle with vertices $A(1,1,2)$, $B(2,3,5)$ and $C(1,5,5)$.	3
12.	Find the projection of the vector $(\vec{b} + \vec{c})$ on the vector $\vec{a}$, where $\vec{a} = 2\hat{\imath} + 2\hat{\jmath} + \hat{k}$, $\vec{b} = \hat{\imath} + 3\hat{\jmath} + \hat{k}$ and $\vec{c} = \hat{\imath} + \hat{k}$.	3
13.	If $\vec{a} = 2\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$, $\vec{b} = -\hat{\imath} + 2\hat{\jmath} + \hat{k}$ and $\vec{c} = 3\hat{\imath} + \hat{\jmath}$, find λ such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$.	2
14.	Find the position vector of the point R which divides the line joining $P(2\vec{a} + \vec{b})$ and $Q(\vec{a} - 3\vec{b})$ externally in the ratio 1: 2.	2
	LEVEL 3 **Q 15 – Q 20**	**23**
15.	If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$, prove that $\vec{a} = \pm 2(\vec{b} \times \vec{c})$.	3
16.	Let $\vec{a} = \hat{\imath} + 4\hat{\jmath} + 2\hat{k}$, $\vec{b} = 3\hat{\imath} - 2\hat{\jmath} + 7\hat{k}$ and $\vec{c} = 2\hat{\imath} - \hat{\jmath} + 4\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 15$.	5
17.	Find a vector of magnitude 5 units perpendicular to each of $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$ and $\vec{b} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$.	3
18.	Using vectors, find the area of the triangle ABC with vertices $A(1,2,3)$, $B(2,-1,4)$ and $C(4,5,-1)$.	3
19.	Show that the points with position vectors $2\hat{\imath} - \hat{\jmath} + \hat{k}$, $\hat{\imath} - 3\hat{\jmath} - 5\hat{k}$ and $3\hat{\imath} - 4\hat{\jmath} - 4\hat{k}$ are the vertices of a right-angled triangle. Also find the remaining angles' cosines.	5

Q No.	Question	Marks
20.	**Case Study:** A toy drone starts from the origin O. It first moves to point A with $\overrightarrow{OA} = 2\hat{\imath} + 3\hat{\jmath} + \hat{k}$ and is then commanded to point B with $\overrightarrow{OB} = 4\hat{\imath} + \hat{\jmath} + 3\hat{k}$ (units in metres). **(i)** Find the displacement vector $\overrightarrow{AB}$. **[1]** **(ii)** Find the distance travelled from A to B and the unit vector along $\overrightarrow{AB}$. **[1]** **(iii)** Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 10: Vector Algebra

Q No.	Answer / Key Steps
	LEVEL 1
1.	(A) $(\hat{\imath}-\hat{\jmath})\times(\hat{\imath}+\hat{\jmath})=2\hat{k}$; unit vector $\hat{k}$
2.	(C) $37=9+16+2\vec{b}\cdot\vec{c}\Rightarrow\vec{b}\cdot\vec{c}=6$, $\cos\theta=\frac{6}{12}=\frac{1}{2}$
3.	(A) $2-2\lambda+3=0\Rightarrow\lambda=\frac{5}{2}$
4.	(C) $\sqrt{100-64}=6$
5.	(B) $\frac{2+6+2}{\sqrt{6}}=\frac{10}{\sqrt{6}}=\frac{5\sqrt{6}}{3}$
6.	(A) $\vec{b}=2\vec{a}$; R explains A.
7.	$\vec{a}\times\vec{b}=-2\hat{\jmath}-2\hat{k}$; area $=2\sqrt{2}$ sq units
	LEVEL 2
8.	$\overrightarrow{AB}=6\hat{\imath}-2\hat{\jmath}-3\hat{k}$, $\lvert\overrightarrow{AB}\rvert=7$; required vector $=-\frac{21}{7}(6\hat{\imath}-2\hat{\jmath}-3\hat{k})=-18\hat{\imath}+6\hat{\jmath}+9\hat{k}$
9.	$\vec{a}\times\vec{b}=\hat{\imath}-2\hat{\jmath}+\hat{k}$; unit vector $\pm\frac{1}{\sqrt{6}}(\hat{\imath}-2\hat{\jmath}+\hat{k})$
10.	$\vec{a}\cdot\vec{b}=12$, $\lvert\vec{a}\rvert=\sqrt{14}$, $\lvert\vec{b}\rvert=\sqrt{24}$; $\cos\theta=\frac{12}{\sqrt{336}}=\frac{\sqrt{21}}{7}$; $\theta=\cos^{-1}\frac{\sqrt{21}}{7}$
11.	$\overrightarrow{AB}\times\overrightarrow{AC}=-6\hat{\imath}-3\hat{\jmath}+4\hat{k}$; area $=\frac{\sqrt{61}}{2}$ sq units
12.	$\vec{b}+\vec{c}=2\hat{\imath}+3\hat{\jmath}+2\hat{k}$; projection $=\frac{4+6+2}{3}=4$
13.	$(2-\lambda)3+(2+2\lambda)=8-\lambda=0\Rightarrow\lambda=8$
14.	$\vec{r}=\frac{1(\vec{a}-3\vec{b})-2(2\vec{a}+\vec{b})}{1-2}=3\vec{a}+5\vec{b}$
	LEVEL 3
15.	$\vec{a}\perp\vec{b},\vec{c}\Rightarrow\vec{a}=\lambda(\vec{b}\times\vec{c})$; $\lvert\vec{b}\times\vec{c}\rvert=\sin\frac{\pi}{6}=\frac{1}{2}$, so $1=\frac{\lvert\lambda\rvert}{2}\Rightarrow\lambda=\pm2$
16.	$\vec{a}\times\vec{b}=32\hat{\imath}-\hat{\jmath}-14\hat{k}$; $\vec{d}=\lambda(32\hat{\imath}-\hat{\jmath}-14\hat{k})$, $9\lambda=15$; $\vec{d}=\frac{1}{3}(160\hat{\imath}-5\hat{\jmath}-70\hat{k})$
17.	$(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})=-2\hat{\imath}+4\hat{\jmath}-2\hat{k}$; required vector $\pm\frac{5}{\sqrt{6}}(-\hat{\imath}+2\hat{\jmath}-\hat{k})$
18.	$\overrightarrow{AB}\times\overrightarrow{AC}=9\hat{\imath}+7\hat{\jmath}+12\hat{k}$; area $=\frac{\sqrt{274}}{2}$ sq units
19.	$\overrightarrow{BC}=2\hat{\imath}-\hat{\jmath}+\hat{k}$, $\overrightarrow{CA}=-\hat{\imath}+3\hat{\jmath}+5\hat{k}$; $\overrightarrow{BC}\cdot\overrightarrow{CA}=0$, right angle at C. $\cos A=\sqrt{\frac{35}{41}}$, $\cos B=\sqrt{\frac{6}{41}}$.
20.	(i) $\overrightarrow{AB}=2\hat{\imath}-2\hat{\jmath}+2\hat{k}$. (ii) $2\sqrt{3}$ m; $\frac{1}{\sqrt{3}}(\hat{\imath}-\hat{\jmath}+\hat{k})$. (iii) $\cos\theta=\frac{14}{\sqrt{14}\sqrt{26}}=\sqrt{\frac{7}{13}}$, $\theta=\cos^{-1}\sqrt{\frac{7}{13}}$